

Biology

Q:1 (MCQs)

1. C

2. B

3. D

4. C

5. C

6. C

7. B

8. B

9. B

10. B

11. C

12. C

Section - B

Q.i(2)

A.i(2)

Biology has 3 main divisions :-

Botany

Zoology

Microbiology

Botany - The study of plants like photosynthesis

Zoology - The study of animals like reproduction of animal.

Microbiology - Microorganism the study of microorganisms like amoeba.

Q.ii)(or)(2)

A.:(ii)(or)(2)

Ronald Ross is a British physician, in 1897 who proved that the malaria is caused by plasmodium and is transmitted in humans through the bite of female mosquito anopheles. He demonstrated the parasite life cycle inside mosquito guts. This discovery established the concept of vector (an organism that transmits ~~the~~ a pathogen) and earned Ross a Nobel Prize in medicine in 1902.

2Q. (iii) (or)

2A. (iii) (or)

In biology, Domain means largest of all groups in the classification of life. Domain is group of Kingdom or taxonomic category above the Kingdom. In 1990, Carl Woese introduced a three domain system of classification of life.

The three domain systems are :-

- ① Archaea
- ② Bacteria
- ③ Eukarya.

Classification into a three domain system is based on the difference of the sequence of nucleotides ~~of~~ in the rRNA (ribosomal ribonucleic acid) of the cell, cell membrane lipid structure and its sensitivity to antibiotics.

① Domain Archaea :-

Following are the characteristics of Domain Archaea :-

- ① Domain Archaea are prokaryotic cells.
- ② Their cell wall contains no peptidoglycan.
- ③ Their rRNA are ~~not~~ not found in Bacteria and Eukarya.

② Domain Bacteria :-

Following are the characteristics of domain Bacteria are :-

- ① Domain Bacteria are prokaryotic cells.
- ② Their cell wall contains peptidoglycan.
- ③ Their rRNA is unique to bacteria.

③ Domain Eukarya :-

Following are the characteristics of domain Eukarya :-

- ① Eukarya have eukaryotic cells.
- ② Not all eukarya have cells in cell wall. Their cell wall contains no peptidoglycan.
- ③ Eukarya rRNA that is unique to eukarya.

Q.2 (iv) (or)

A.2 (iv) (or)

Rough endoplasmic reticulum	Smooth endoplasmic reticulum
① It is present in ribosome and it gives rough appearance.	It is absent in ribosome.
② It's function is protein synthesis and transport.	It's function is lipid synthesis and detoxification.
③ It's present in ^{near} nucleus.	It present in cytoplasm.

Q: 2(v)
A: 2(v)

• Genetic stability:

Mitosis produces 2 daughter cells with the same number of chromosomes as in parent cell. Before the start of cell division, DNA of parental chromosomes to form its 2 exact copies. Daughter cells thus will have same genetic information.

• Development and growth :-

Life of sexually reproducing multicellular organism starts from a single cell, the zygote. The organisms also grow in size by increasing cell number. The development or growth of multicellular organisms depends on mitosis.

Q: 2(vi)(or)
A: 2(vi)(or)

Cancer is a disease of cell cycle. ~~As~~ Unlike all normal cells ~~off cycle~~ of the body ~~cells~~, cancer cells do not have proper functioning cell cycle control system and therefore divides excessively. This excessive growth can result in an ~~abnormal~~ abnormal mass of cells called tumour. However, a benign tumour is an abnormal mass of essential normal cells. They

They always remain at their original site on the body. A malignant tumour is cancerous. It is capable of spreading into neighbouring tissues and often to other distant parts of the body. The spread of cancer cells beyond their original site is called metastasis.

Q.2(vii)
A.2(vii)

• Epithelial tissue :-

Inner lining, secretes mucus, HCl and pepsin enzyme

1 • Connective tissue :-

Supports and connects the mucosa to the muscular layer.

• Muscle tissue :-

Contracts to move food through the gut.

• Nervous tissue :-

Controls smooth muscle contraction.

Q.2(viii)

A.2(viii)

Plant tissues differentiate into three main types :-

(P.T.O)

Dermal tissue :-
covers and protects the plant.

Ground tissue :-
serves as ^{for} site in photosynthesis.

vascular tissues :-
helps to support transports
water ~~and~~ ^{minerals} and sugars to different
parts of plants.

S: $2(n)$
A: $2(n)$

mRNA :-
It carries message of DNA from nucleus
to cytoplasm for protein synthesis so called
mRNA (messenger ribonucleic acid).

tRNA :-
It's transfer amino acids to ribosomes
during protein synthesis, so called ~~TRA~~ tRNA.
(transfer ribonucleic acid)

rRNA :-
It is the component of ribosomes
therefore called rRNA (ribosomal ribonucleic
acid). It combines with ribosomal proteins
to form units of ribosomes which are
factory of proteins.

Q: 2 (2i)

A: 2 (2i)

Four factors are :-

- ① Temperature
- ② pH
- ③ substrate concentration
- ④ Inhibitors / Activators.

Temperature:

The rate of reaction increases with a increase in temperature but upto a specific limit. Each work best at a specific temperature called optimum temperature. Optimum temperature for human ~~enzymes~~ enzymes is 37°C .

At the very high temperature atoms of enzyme molecule start vibrating vigorously so globular structure and active site of enzymes is damaged and it is called denatured.

Section - C

Q: 3(i)(or)

A: 3(i)(or)

Advantages of classification:-

- Makes study of organisms easy and systematic
- Helps understand evolutionary relationships.
- Provides a universal language for scientists.
- Facilitates identification of new species and organisms.

~~Tax~~ Taxonomic hierarchy :-

Taxa	Human
Domain	Eukarya
Kingdom	Animalia
Phylum	Chordata
Class	Mammalia
Order	Primates
Family	Hominidae
Sp. Genus	Homo
Species	sapiens sapiens.

Q: 3 (ii)

A: 3 (ii)

a) Mitochondria:

Structure:-

Double membrane organelle. Outer membrane is smooth, inner membrane is folded into projections called cristae. The fluid inside is called matrix. It contains its own DNA and ribosomes.

Functions:-

- Site of cellular respiration; produces ATP (energy currency). It is called "powerhouse of the house". One glucose molecule yields 36 ATP in aerobic respiration.

b) Golgi Apparatus:-

Structure:-

- Stack of flattened membrane called sacs called cisternae, discovered by Camillo golgi.

Functions:-

Receives proteins from RER, processes/modifies and packages them into vesicles for secretion or delivery to lysosomes.

Q: 3 (iii) (or)
A: 3 (iii) (or)

The flow of genetic information from DNA to RNA to proteins is called the central dogma of ~~life~~ molecular biology.

★ Transcription :-

- First step of gene expression occurring in nucleus. Both strands of a gene unwind, only one strand (the coding strand) is copied. The nucleotide sequence of the coding DNA strand is copied as mRNA. After copying both DNA strands re-coil into the double helix. mRNA then moves from nucleus to cytoplasm.

★ Translation :-

- It occurs at ribosomes in the cytoplasm. mRNA codons are read by ~~ribon~~ ribosomes, each codon specifies one amino acid. tRNA molecules carry specific amino acid to ribosomes, matching anticodon to codon.

Q.3(iv)(or)
Ans.3(iv)(or)

ATP as energy (or) currency :-
All life processes (movement, development, reproduction etc) depend on energy. Living organisms store and use energy at the cellular level as Adenosine triphosphate (ATP). ATP is called energy currency of the cell. It stores and releases energy like money. It can be used for any work.

Structure of ATP :-

- Adenine (double-ringed nitrogen base)
- Ribose (five-carbon sugar).
- Adenine + Ribose = Adenosine
- Three Phosphate group (PO_4) linked in a chain called a triphosphate group.

Adenosine + 1P =
AMP

Adenosine + 2P =
ADP

Adenosine + 3P =
ATP

High energy bonds :-

The covalent bonds b/w phosphate groups are called high-energy bonds. Represented by ($\sim$). These bonds store large amounts of energy that cells use to carry out their functions.